

Chapter 3 Extension Polynomial and Rational Inequalities Exercises

Directions: In exercises 1 – 20, solve each inequality and write the solution in interval notation.

1. $x^2 + 3x - 4 \geq 0$

2. $x^2 + x - 6 \leq 0$

3. $x^2 - 7x + 10 < 0$

4. $x^2 - 4x + 3 > 0$

5. $x^2 + 8x > -15$

6. $x^2 + 8x < -12$

7. $x^2 - 4x + 2 \leq 0$

8. $-x^2 + 8x - 11 < 0$

9. $x^2 - 10x > -19$

10. $x^2 + 6x < -3$

11. $-6x^2 + 19x - 10 \geq 0$

12. $-3x^2 - 4x + 4 \leq 0$

13. $-2x^2 + 7x + 4 \geq 0$

14. $2x^2 + 5x - 12 > 0$

15. $x^2 + 3x + 5 > 0$

16. $x^2 - 3x + 6 \leq 0$

17. $-x^2 + x - 7 > 0$

18. $-x^2 - 4x - 5 < 0$

19. $-2x^2 + 8x - 10 < 0$

20. $-x^2 + 2x - 7 \geq 0$

Directions: In exercises 21 - 52, solve each inequality and write the solution in interval notation.

21. $\frac{x-3}{x+4} \geq 0$

22. $\frac{x+6}{x-5} \geq 0$

23. $\frac{x+1}{x-3} \leq 0$

24. $\frac{x-4}{x+2} \leq 0$

25. $\frac{x-7}{x-1} > 0$

26. $\frac{x+8}{x+3} > 0$

$$27. \frac{x-6}{x+5} < 0$$

$$28. \frac{x+5}{x-2} < 0$$

$$29. \frac{3x}{x-5} < 1$$

$$30. \frac{5x}{x-2} < 1$$

$$31. \frac{6x}{x-6} > 2$$

$$32. \frac{3x}{x-4} > 2$$

$$33. \frac{2x+3}{x-6} \leq 1$$

$$34. \frac{4x-1}{x-4} \leq 1$$

$$35. \frac{3x-2}{x-4} \geq 2$$

$$36. \frac{4x-3}{x-3} \geq 2$$

$$37. \frac{1}{x^2+7x+12} > 0$$

$$38. \frac{1}{x^2-4x-12} > 0$$

$$39. \frac{3}{x^2-5x+4} < 0$$

$$40. \frac{4}{x^2+7x+12} < 0$$

$$41. \frac{2}{2x^2+x-15} \geq 0$$

$$42. \frac{6}{3x^2-2x-5} \geq 0$$

$$43. \frac{-2}{6x^2-13x+6} \leq 0$$

$$44. \frac{-1}{10x^2+11x-6} \leq 0$$

$$45. \frac{1}{2} + \frac{12}{x^2} > \frac{5}{x}$$

$$46. \frac{1}{3} + \frac{1}{x^2} > \frac{4}{3x}$$

$$47. \frac{1}{2} - \frac{4}{x^2} \leq \frac{1}{x}$$

$$48. \frac{1}{2} - \frac{3}{2x^2} \geq \frac{1}{x}$$

$$49. \frac{1}{x^2-16} < 0$$

$$50. \frac{4}{x^2-25} > 0$$

$$51. \frac{4}{x-2} \geq \frac{3}{x+1}$$

$$52. \frac{5}{x-1} \leq \frac{4}{x+2}$$